FARJANA AKTHER KOLY



| Oceanography Graduate | Participatory Research | Project Documentation | GIS/RS |

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LinkedIn | Portfolio: https://farjanaaktherkoly.github.io/ | ResearchGate

PROFESSIONAL SUMMARY

Oceanography graduate and M.Sc. thesis student with experience in research coordination, GIS/Remote Sensing, quantitative and qualitative data analysis, field research, project documentation, and scientific writing. Currently coordinating B4RL Research Wing activities and supporting undergraduate and master's students in research design, data collection, analysis, analysis, thesis writing, and manuscript preparation. Proficient in ArcGIS Pro, QGIS, Python, R, SPSS, and MS Excel, with strong skills in stakeholder coordination, technical reporting, and interdisciplinary environmental research.

WORK EXPERIENCES

Research Coordinator | **Centre for Climate Change and Disaster Resilience (CDR), North South University**

August 2026–Present | Full-Time

• Coordinate B4RL Research Wing activities, stakeholder communication, fieldwork, surveys, workshops, presentations, and progress reporting.

• Support undergraduate and master's thesis students in research design, data collection and analysis, thesis writing, and manuscript preparation.

Research Associate | **Bangladesh Maritime University** | October 2025–August 2026 | Full-Time

• Support climate, coastal, and environmental research through data analysis, GIS/RS, literature synthesis, technical reporting, and research outputs.

• Assist manuscript and proposal development, including methods, results interpretation, figures/tables, and scientific writing.

Researcher | **Hydro-Climate & Ocean Centre**

January 2025–Present | Part-Time

• Support climate and coastal research through environmental data analysis, fieldwork, project documentation, and technical writing on vulnerability, risk, water quality, and resilience.

Researcher | **National Oceanographic and Maritime Institute**

January 2023–December 2024 | Part-Time

• Supported field and desk-based environmental research, data collection, reporting, and multidisciplinary research communication.

EDUCATION

M.Sc. in Oceanography

Bangladesh Maritime University

Ongoing; | Current M.Sc. Thesis Semester

B.Sc. in Oceanography

Bangladesh Maritime University

2023 | CGPA 3.87/4.00

HSC and SSC (Science)

Sylhet Board

HSC GPA 5.00/5.00 |

SSC GPA 5.00/5.00

SKILLS

Remote Sensing & GIS

ArcGIS Pro | **ArcGIS** | Google Earth Engine | field validation

Data & Analytics

Python | R | **Power BI** | MS Excel | SPSS | Origin Pro | descriptive statistics | data cleaning & validation

Research & Writing

Evidence synthesis | technical writing | report preparation |

Project & Field Support

Data Collection & survey support

Instruments handling

CTD, Multi-meter, Algal Torch, Spectrophotometer, AAS, etc. |

LANGUAGES

Bangla: Native/Fluent;

English: Excellent

PROJECTS EXPERIENCES

1. **Youth4Climate Coastal Mangrove Restoration and Monitoring:** GIS/RS-based mangrove restoration planning, field validation, monitoring documentation and project coordination.
2. **Climate-Induced PFZ Shifts in Bangladesh EEZ:** environmental data handling, GIS/RS interpretation, documentation and climate-related livelihood risk analysis.
3. **Groundwater Quality Assessment in South-Western Bangladesh:** hydrogeochemical data organization, analysis support, figures, tables and technical documentation.
4. **Sundarbans Ecosystem Health Assessment:** environmental indicator interpretation, climate-vulnerability context, documentation and research output preparation.
5. **River Ecosystem Health Assessment of Bangladesh:** water-quality data handling, analysis support, technical documentation and reporting.

AWARDS AND LEADERSHIP

- **Vice Chancellor Scholarship** for outstanding academic performance in B.Sc. (2022, 2023, 2024).
- **NST Fellowship 2025-26**, Ministry of Science and Technology, Government of Bangladesh.
- Supported student research, documentation, and collaborative research communication across university and research networks.

INTERESTED FIELDS

Remote sensing, GIS, Data Analysis, environmental studies, water resources & modelling, oceanography, blue economy, coastal resilience, mangrove restoration, climate vulnerability, etc.

PUBLICATIONS AND MANUSCRIPTS

- Choudhury, T. R., Alam, M. A., Sowrav, S. F. F., Koly, F. A., & Chowdhury, A. S. (2026). Tracing the sources and environmental distribution of PFAS and heavy metals in the Bakkhali River and coastal zone of Cox's Bazar, Bangladesh. Journal of Contaminant Hydrology, 282, 105026. https://doi.org/10.1016/j.jconhyd.2026.105026
- Sunny, S. K., Koly, F. A., Ahammed, A. A., & Firdaws, S. J. (2026). Assessment of groundwater quality and its implications for sustainable water management in southwestern Bangladesh. Bangladesh Maritime Journal, 10(1), 91-107. DOI: 10.70279/bmj-v1-1097
- Hasan, Md. A., et al. (2026). Assessing the dynamics of water quality and their implications on biochemical health of river ecosystem: A RS-based study of the Shitalakshya River, Dhaka. Bangladesh Maritime Journal, 10(1), 313-329. DOI: 10.70279/bmj-v2-1124
- **Manuscripts under review/in progress:** Climate Change Impacts on Coastal Chattogram; Marine Spatial Planning and Zoning for Sustainable Blue Economy in Bangladesh EEZ; OMZ Prediction using ML; MHW Exposure using ML/DL; PFAS; Dhaka River Ecosystem Dynamics.

TRAINING AND CERTIFICATIONS

- Coastal and Marine Ecosystem Protection for a Sustainable Blue Economy in Bangladesh
  – IOI Ocean Academy Bangladesh, 2026.
- Python Programming: Zero to Expert in 40 Days
  – DSD Research Firm, 2026.
- Coastal and Regional Ocean Community (CROCO) Model – BMU, 2025.

REFERENCES

Dr. S. M. Mustafizur Rahman | Assistant Professor, Bangladesh Maritime University | mustafizur.ocn@bmu.edu.bd | +8801401807894

Dr. Mohan Kumar Das | Executive Director, Hydro-Climate & Ocean Center (HCO) | mohan.smrc@gmail.com | +8801913630710

Sheikh Fahim Faysal Sowrav | Lecturer, Bangladesh Maritime University | sowrav.ocean@gmail.com | +8801986761741